

CHAPTER 01

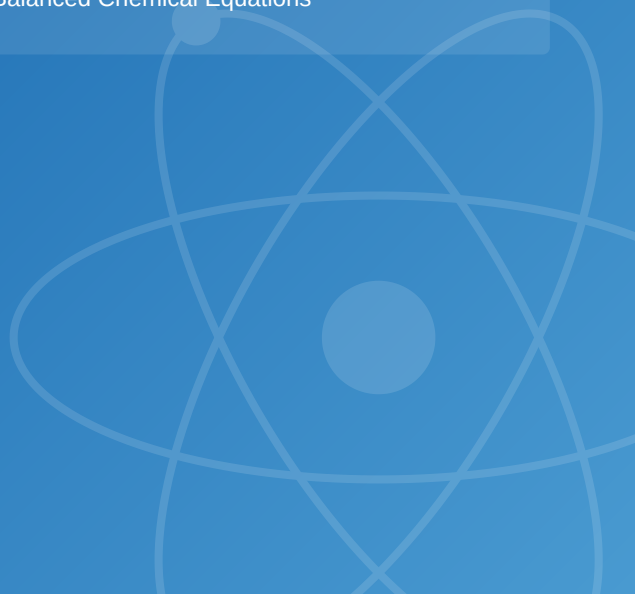
Matter, Atoms & The Periodic Table

From the screen you're reading this on to the air filling your lungs — everything in the universe is built from the same tiny LEGO® bricks. Time to learn what they are and how scientists organized them all.

 Georgia Standard S8P1

WHAT YOU'LL LEARN

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|--|--|
| 01 What Is Matter? | 06 Inside the Atom |
| 02 Classifying Matter | 07 Reading the Periodic Table |
| 03 Physical vs. Chemical Properties | 08 Patterns in the Periodic Table |
| 04 States of Matter | 09 Counting Atoms in Formulas |
| 05 Heating & Cooling Curves | 10 Balanced Chemical Equations |



1.1 What Is Matter?

Look around you. Your phone, your desk, the air you're breathing, even *you* — it's all matter. Scientists define **matter** as anything that has **mass** and takes up **space** (volume). That definition sounds simple, but it covers *almost everything in the universe*.

If something has mass and volume, it's matter. A water bottle? Matter. A cloud? Matter. The wind blowing your hat off? Yep, matter — air has mass too, even if you can't see it. The Wi-Fi signal letting you scroll TikTok? That's *not* matter — it's energy, and we'll get to that in Chapter 2.

! Did You Know?

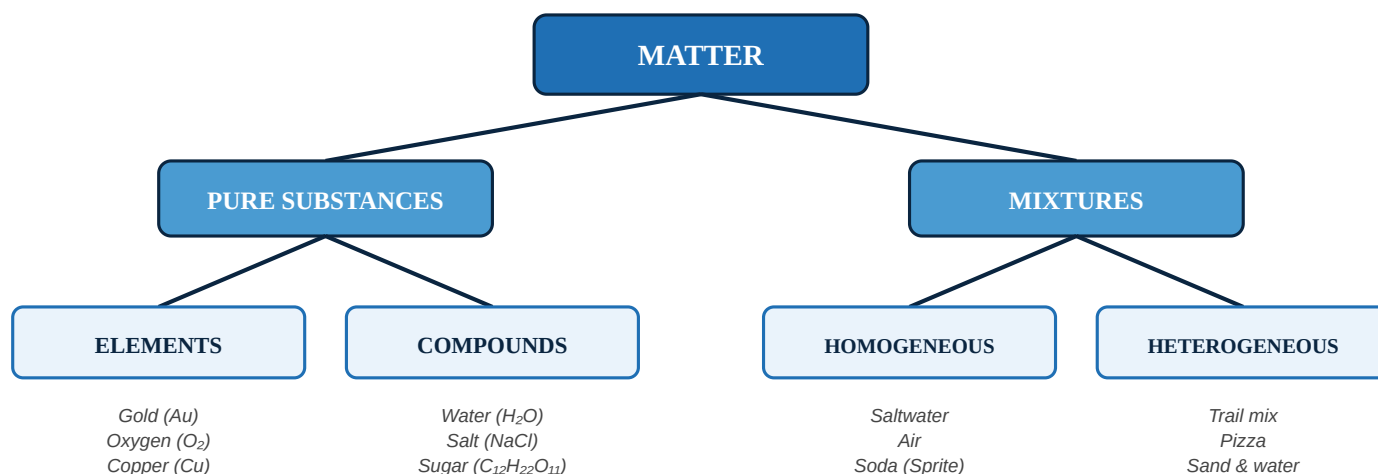
Every atom in your body was forged inside a star that exploded billions of years ago. The carbon in your hair, the calcium in your bones, the iron in your blood — all of it is literally **stardust**. Astronomer Carl Sagan put it best: "We are made of star-stuff."

X Common Mix-Up

Mass ≠ Weight. Mass is how much "stuff" is in an object — it never changes. Weight is the pull of gravity on that mass. You'd have the same mass on the Moon, but you'd weigh about 1/6 as much because the Moon's gravity is weaker.

1.2 Classifying Matter

All matter can be sorted into two huge buckets: **pure substances** and **mixtures**. Each one then splits again. Think of it like sorting your music library — you start broad, then get more specific.



Same throughout (looks uniform) vs. Different parts visible

Figure 1.1 — The classification of matter. Every type of matter fits somewhere on this chart.

Pure Substances: Elements & Compounds

A **pure substance** has only one kind of "stuff" in it — its makeup is the same all the way through.

An **element** is the simplest type of matter. It's made of only one kind of atom and can't be broken down by ordinary chemistry. Gold is just gold atoms. Helium is just helium atoms. There are about 118 known elements, and they all live on the periodic table (we'll meet it soon).

A **compound** is two or more different elements **chemically bonded** together in a fixed ratio. Water (H₂O) is always two hydrogen atoms stuck to one oxygen atom — every single time. Compounds have totally different properties than the elements that make them up. Sodium is a soft metal that explodes in water. Chlorine is a poisonous green gas. Bond them together? You get sodium chloride — regular table salt that you put on your fries.

Mixtures: Homogeneous & Heterogeneous

A **mixture** is two or more substances **physically** combined — they're hanging out together but not chemically bonded. That means you can separate them again (sometimes easily, sometimes not).

A **homogeneous mixture** looks the same throughout — you can't see the different parts. Saltwater looks like plain water, but the salt is dissolved in it. We also call these **solutions**. A **heterogeneous mixture** has visibly different parts. Look at a slice of pepperoni pizza: you can see the cheese, the sauce, the pepperoni. Different parts, mixed together.

? Quick Check 1.2

1. Classify each as element, compound, homogeneous mixture, or heterogeneous mixture: (a) a bowl of cereal with milk, (b) pure copper wire, (c) CO₂, (d) apple juice.
2. Why is salt (NaCl) a compound and not a mixture?

1.3 Physical vs. Chemical Properties

How do scientists tell substances apart? They look at **properties** — the characteristics of matter. There are two kinds.

Physical Properties

A **physical property** is something you can observe or measure **without changing what the substance is**. You can measure how heavy a brick is, or how shiny a penny is, and the brick is still a brick and the penny is still a penny.

Physical Property	What It Means	Example
Density	How much mass fits in a certain volume ($D = m/V$)	Lead is dense; styrofoam isn't
Melting Point	Temperature where a solid turns to liquid	Ice melts at 0°C
Boiling Point	Temperature where a liquid turns to gas	Water boils at 100°C
Color, Shine, Texture	What you see and feel	Gold is yellow and shiny
Solubility	Ability to dissolve in something else	Sugar dissolves in water
Magnetism	Attracted to magnets or not	Iron is magnetic; aluminum isn't

Chemical Properties

A **chemical property** describes **how a substance reacts with other substances**. To observe these, you usually have to change the substance into something new.

Chemical Property	What It Means	Example
Reactivity	How easily a substance reacts with others	Sodium reacts violently with water
Combustibility	The ability to burn	Gasoline is highly combustible
Flammability	How easily it catches fire	Paper burns; rock doesn't
Toxicity	Ability to harm living things	Mercury is toxic to humans
Tarnishing/Rusting	Reacting with oxygen in air	Iron rusts; gold doesn't

Physical vs. Chemical Changes

Properties lead to **changes**. A **physical change** changes form or appearance, but the substance is **still the same stuff**. Cutting paper, melting ice, dissolving sugar in tea — all physical. A **chemical change** produces an entirely **new substance** with new properties. Burning toast, rusting metal, baking a cake — all chemical.

★ Signs of a Chemical Change

How do you know a chemical change just happened? Look for these clues: **(1)** color change that doesn't reverse, **(2)** a gas given off (bubbles or smell), **(3)** a solid forming from liquids (called a precipitate), **(4)** light or heat released, **(5)** the change is hard to reverse. Spotting two or more? Probably chemical.

✗ Common Mix-Up

Dissolving is physical, not chemical. When you stir sugar into iced tea, the sugar disappears — but it's still sugar! You could evaporate the water and get the sugar back. No new substance formed. Now *digesting* that sugar? That's chemical — your body breaks it down into different molecules.

? Quick Check 1.3

1. Label each as physical or chemical change: (a) a banana turning brown, (b) shredding paper, (c) water freezing, (d) fireworks exploding, (e) chopping wood, (f) burning wood.
2. Is "boiling point" a physical or chemical property? Defend your answer.

1.4 States of Matter

Matter exists in four common **states** (also called phases): solid, liquid, gas, and plasma. The difference between them isn't *what* the matter is — it's **how its particles are moving**.

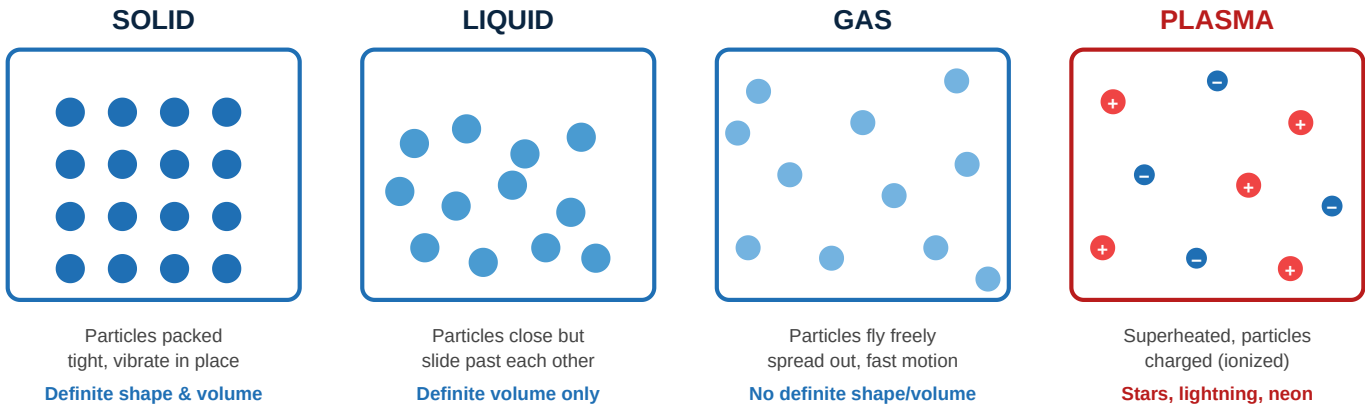


Figure 1.2 — The four states of matter. The difference is all about particle motion.

When you **add thermal energy** (heat) to a substance, its particles move faster and spread out — solids melt to liquids, liquids evaporate to gases. When you **remove thermal energy**, the opposite happens — gases condense, liquids freeze.

! Did You Know?

About **99% of all visible matter in the universe is plasma** — every star, including our Sun, is a giant ball of plasma. Down here on Earth, plasma is rare, but you've seen it: lightning, the flame on a gas stove, and the gas inside a neon sign are all plasma.

1.5 Heating & Cooling Curves

If you put a chunk of ice on the stove and watched a thermometer carefully, you'd notice something surprising: the temperature doesn't just go up steadily. It **pauses** during phase changes. A **heating curve** shows this on a graph.

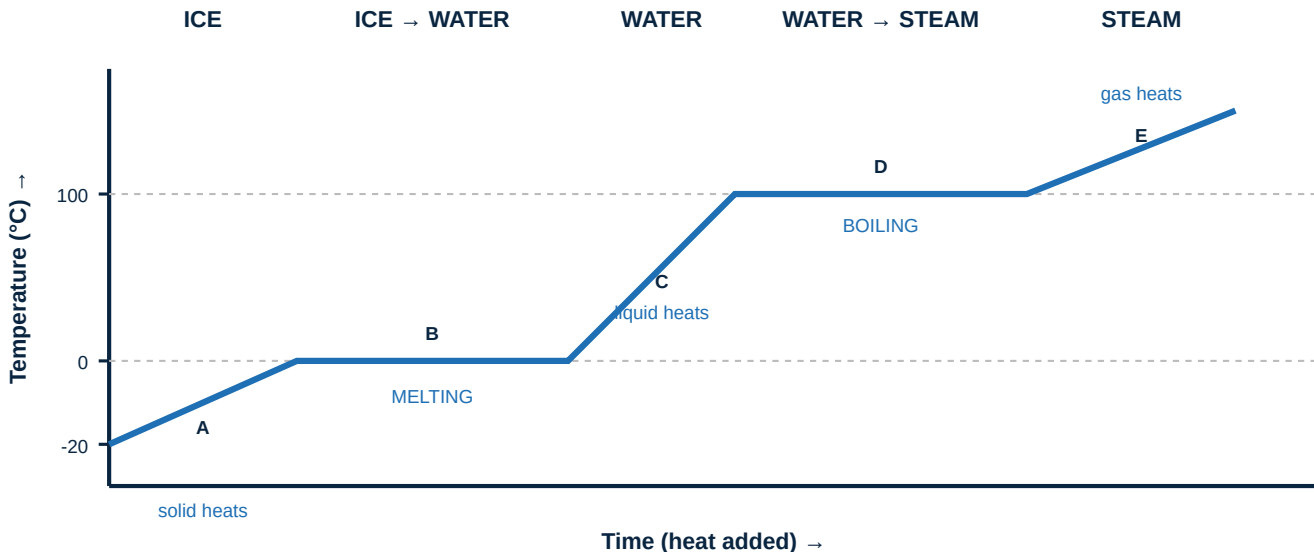


Figure 1.3 — Heating curve for water. Notice the flat spots during phase changes (B and D).

During the **sloped sections** (A, C, E), heat is being added and temperature is rising. The particles are moving faster. During the **flat sections** (B, D), heat is still being added — but the temperature isn't changing! That's because all the energy is being used to **break the bonds** between particles so the substance can change state. Once the phase change is complete, temperature starts rising again.

A **cooling curve** is the same idea in reverse — you'd see flat spots at the freezing point and the condensation point as the substance loses energy and transitions back down through the states.

? Quick Check 1.5

1. On a heating curve, what is happening to the particles during a flat (horizontal) section?
2. If you removed thermal energy from steam at 105°C, predict what would happen at 100°C and again at 0°C.

1.6 Inside the Atom

Now we zoom in. WAY in. Take any element — say, a piece of copper — and keep cutting it in half. Eventually you'll reach a piece so small that cutting it again would change what it is. That tiniest piece is an **atom**.

But atoms aren't the smallest things. Each atom is built from three **subatomic particles**: protons, neutrons, and electrons.

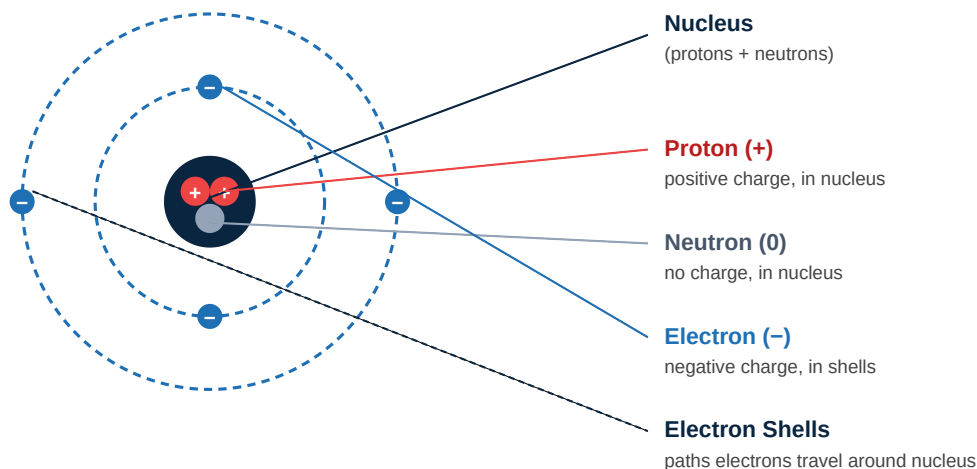


Figure 1.4 — A model of a helium atom. The nucleus is tiny but holds almost all the mass.

Particle	Charge	Location	Relative Mass
Proton	+1 (positive)	Nucleus	~1 amu
Neutron	0 (neutral)	Nucleus	~1 amu
Electron	-1 (negative)	Shells around nucleus	~1/2000 amu (basically nothing)

The **nucleus** is incredibly dense — protons and neutrons packed together in the center. Electrons zip around it in regions called **shells** or **energy levels**. Most of an atom is actually **empty space**. If an atom were the size of a football stadium, the nucleus would be the size of a pea sitting on the 50-yard line!

! Did You Know?

Atoms are so small that a single drop of water contains about **1.5 sextillion molecules** (that's 1,500,000,000,000,000,000,000). Each one is made of three atoms. If you stacked atoms on top of each other, it would take 10 million of them to equal the thickness of a human hair.

Identifying an Atom

How do scientists tell atoms apart? Two key numbers:

- **Atomic Number** = the number of **protons**. This number is unique for each element. Carbon has 6 protons. Always. If it has 7 protons, it's not carbon anymore — it's nitrogen.
- **Mass Number** = protons + neutrons (electrons are too tiny to count).

In a **neutral atom**, the number of electrons equals the number of protons. The positive and negative charges cancel out.

? Quick Check 1.6

1. An atom has 11 protons, 12 neutrons, and 11 electrons. What is its atomic number? Its mass number? Its overall charge?
2. Which subatomic particle determines what element an atom is?

1.7 Reading the Periodic Table

In 1869, a Russian chemist named **Dmitri Mendeleev** got tired of all the elements being scattered around with no order. He played around with arrangements and discovered that if you organize elements by atomic mass, **patterns repeat**. His chart was so good he could predict elements that hadn't been discovered yet. That chart became the **periodic table**.

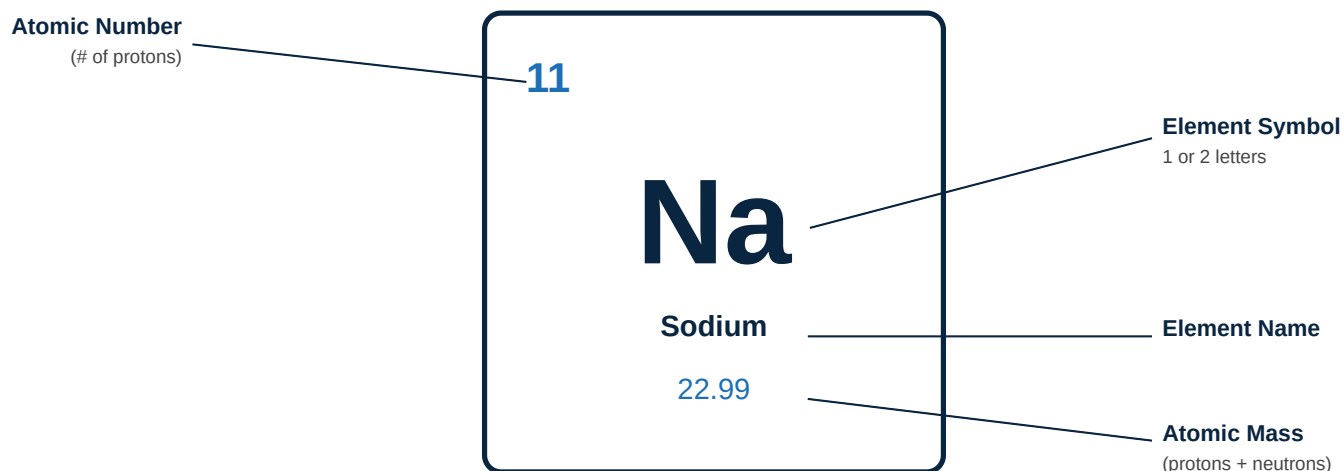


Figure 1.5 — Anatomy of a periodic table square. Every element gives you four key pieces of info.

Finding Subatomic Particles from the Table

★ The Formula Trick

For any element on the periodic table:

- **# of Protons** = atomic number
- **# of Electrons** = atomic number (in a neutral atom)
- **# of Neutrons** = atomic mass (rounded) – atomic number

Example: Sodium has an atomic number of 11 and an atomic mass of about 23. So it has 11 protons, 11 electrons, and $23 - 11 = 12$ neutrons.

1.8 Patterns in the Periodic Table

The periodic table isn't a random grid — it's organized into **rows and columns** that tell you about element properties.

Periods (Rows)

A **period** is a **horizontal row**. There are 7 periods. As you move across a period from left to right, the atomic number increases by one each time. Each period also tells you something cool: the period number equals the **number of electron shells** an atom has. Atoms in Period 1 (hydrogen, helium) have one shell. Atoms in Period 3 (sodium, magnesium, etc.) have three shells.

Groups (Columns / Families)

A **group** (also called a **family**) is a **vertical column**. There are 18 groups. Elements in the same group have **similar chemical properties** because they have the same number of electrons in their outermost shell (called **valence electrons**). This is the secret to why the periodic table works!

Group	Name	Properties
1	Alkali Metals	Soft, super reactive, explode in water (not hydrogen)
2	Alkaline Earth Metals	Reactive metals, less so than Group 1
3–12	Transition Metals	Hard, shiny, conduct electricity (iron, copper, gold)
17	Halogens	Very reactive nonmetals (chlorine, fluorine)
18	Noble Gases	Almost zero reactivity — outer shell is "full"

The Big Three Regions

If you draw a "staircase" line from boron down toward astatine on the right side of the table, it splits the elements into:

Metals (left of staircase): shiny, malleable (can be hammered), ductile (can be drawn into wires), good conductors of heat and electricity. About 75% of all elements!

Nonmetals (right of staircase): dull, brittle, poor conductors. Many are gases at room temperature.

Metalloids (along the staircase): silicon, germanium, arsenic — they have a mix of metal and nonmetal properties. Silicon is the reason computer chips exist!

★ Real-World Connection

Why does your phone work? Because of **silicon**, a metalloid. It's a semi-conductor — it can conduct electricity sometimes and block it other times, which is exactly what computer chips need to make 1s and 0s. Silicon Valley got its name from this element.

? Quick Check 1.8

1. Lithium (Li) and Sodium (Na) are both in Group 1. Why would you predict they have similar properties?
2. Which group contains the LEAST reactive elements? Why are they so chill?

1.9 Counting Atoms in Chemical Formulas

A **chemical formula** is the "recipe" for a compound. It uses element symbols and tiny numbers called **subscripts** to tell you how many atoms of each element are present.

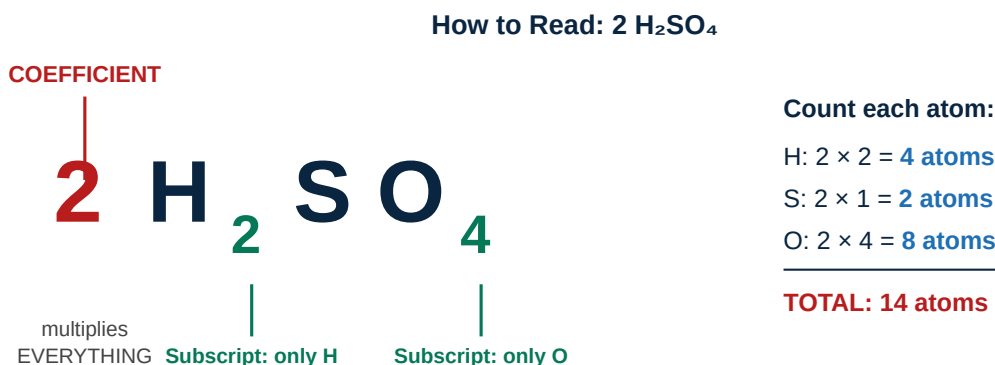


Figure 1.6 — Subscripts apply only to the symbol they follow. Coefficients multiply EVERYTHING in the formula.

★ The Two Rules of Counting Atoms

1. **Subscripts** (small numbers BELOW) only apply to the element symbol directly in front of them. If there's no subscript, assume it's a 1.
2. **Coefficients** (big numbers in FRONT of the formula) multiply every atom in the formula. **3 H₂O** means $3 \times 2 = 6$ hydrogen atoms and $3 \times 1 = 3$ oxygen atoms.

Parentheses

Sometimes a formula has parentheses, like **Ca(NO₃)₂**. The subscript outside the parenthesis applies to **everything inside**. So Ca(NO₃)₂ has 1 calcium, 2 nitrogens (1×2), and 6 oxygens (3×2).

? Quick Check 1.9

1. How many of each atom are in 2 C₆H₁₂O₆ (glucose)?
2. How many of each atom are in Mg(OH)₂?
3. How many TOTAL atoms are in 3 NH₄Cl?

1.10 Identifying Balanced Equations

Time to tie it all together. Remember from Section 1.3 that a **chemical change** creates new substances. We write these changes as **chemical equations**:

Reactants → Products

The stuff you start with (on the left) are **reactants**. The new stuff you end up with (on the right) are **products**. The arrow means "produces" or "yields."

The Law of Conservation of Mass

Here's one of the most important rules in all of chemistry, discovered by Antoine Lavoisier in the 1700s:

! Law of Conservation of Mass

Matter cannot be created or destroyed in a chemical reaction. Atoms just get rearranged — they don't disappear and they don't appear out of nowhere.

This means the **mass of the reactants** must equal the **mass of the products**. Every. Single. Time.

For an equation to be **balanced**, the number of each type of atom on the left has to equal the number of each type of atom on the right. Same atoms, just rearranged.



REACTANTS (Left)

H: 2 atoms
O: 2 atoms

PRODUCTS (Right)

H: 2 atoms ✓
O: 1 atom X



REACTANTS (Left)

H: $2 \times 2 = 4$ atoms
O: 2 atoms

PRODUCTS (Right)

H: $2 \times 2 = 4$ atoms ✓
O: $2 \times 1 = 2$ atoms ✓

Figure 1.7 — In the unbalanced equation, oxygen atoms "vanish" — that violates conservation of mass! Adding coefficients fixes it.

Checking if an Equation is Balanced

★ 3-Step Check

1. List every element that appears in the equation.
2. Count atoms of each element on the LEFT (reactants). Don't forget coefficients!
3. Count atoms of each element on the RIGHT (products). If the numbers match for every element → **BALANCED!**

X Common Mix-Up

To balance an equation, you can **only change coefficients** — the big numbers in front. You can **NEVER change subscripts!** Changing H_2O to H_2O_2 doesn't balance an equation — it changes the substance entirely (H_2O is water, H_2O_2 is hydrogen peroxide).

Why Products Can Look Different From Reactants

When reactants form products, the **atoms are the same** — but they're now bonded differently. That's why the products have totally different properties. Hydrogen is an explosive gas. Oxygen is what we breathe. Combine them in a reaction and you get water — a liquid you can drink. Same atoms, new arrangement, completely new substance.

? Quick Check 1.10

1. Is this equation balanced? $\text{CH}_4 + 2 \text{O}_2 \rightarrow \text{CO}_2 + 2 \text{H}_2\text{O}$ (show your atom counts!)
2. If 50g of reactants go into a sealed reaction, what mass of products must come out? Why?
3. Why can't you balance an equation by changing subscripts?

★ CHAPTER 1 BIG IDEAS

- **Matter** is anything with mass and volume. It comes in pure substances (elements, compounds) and mixtures (homogeneous, heterogeneous).
- **Physical properties** (density, melting point, boiling point) don't change the substance. **Chemical properties** (reactivity, combustibility) describe how it reacts.
- Matter exists as **solid, liquid, gas, and plasma**. Adding/removing thermal energy moves it between states.
- Atoms are made of **protons** (+, in nucleus), **neutrons** (0, in nucleus), and **electrons** (–, in shells).
- The **periodic table** is organized by atomic number. Rows = periods. Columns = groups (similar properties).
- In a chemical formula, **subscripts** apply only to the symbol they follow; **coefficients** multiply everything.
- The **Law of Conservation of Mass** says atoms aren't created or destroyed in reactions — they're rearranged. Balanced equations prove this.

I Can... (Self-Check)

- ☐ I can compare and contrast pure substances and mixtures using a model. (S8P1.a)
- ☐ I can describe how particles move in solids, liquids, gases, and plasma when thermal energy is added or removed. (S8P1.b)
- ☐ I can compare physical properties (density, melting point, boiling point) and chemical properties (reactivity, combustibility) of matter. (S8P1.c)
- ☐ I can classify changes in matter as physical or chemical and defend my answer with evidence. (S8P1.d)
- ☐ I can describe the structure of atoms (protons, neutrons, electrons) and use the periodic table to identify patterns and characteristics. (S8P1.e)
- ☐ I can explain conservation of matter in chemical reactions and describe the differences between products and reactants. (S8P1.f)